

Investigation of early school leaving risk in disadvantaged girl students at primary education level in rural regions by CHAID analysis

Yusuf Kızıltas^{a,*}, Hamza Gönülal^b

^a Van Yüzüncü Yıl University, Faculty of Education, Van, Turkey

^b Ministry of National Education, Van, Turkey

ARTICLE INFO

Keywords:

Rural regions
Primary education teacher
Disadvantaged girl students
Early school leaving
Chaid analysis

ABSTRACT

Disadvantaged girl students in rural areas are often neglected, abused, married at a young age and at risk of early school leaving. The main purpose of this study was to reveal the early school leaving risks for girl students attending primary education in rural areas of east in Türkiye. In the study, which was carried out using the relational research model, a quantitative research method, girl students attending primary education constituted the sample of the study ($n = 515$). Variables explaining the early school leaving risks of girl students were determined using CHAID analysis, a decision tree method. According to the research results, the high number of siblings in girls attending primary education is a dominant variable in the risk of early school leaving.

1. Introduction

Most girls in rural areas face difficulties in their lives. Inadequate and weak educational programs can cause these difficulties (Gulma & Lawal, 2010). During the education process; Early marriage, being busy with housework, supporting the care and education of their siblings, lack of interest in classes, having to work in income-generating activities, and physical difficulties are also the difficulties experienced by girls. These difficulties can lead to dropping out of school (Bashir et al., 2014). It is also an important problem that girls are not given as much importance as boys. Families' indifference towards their daughters centers around health, emotionality and education issues. Girls whose access to education is restricted face negative parental attitudes (Radiowala & Molwane, 2021). These problems experienced in rural areas can put girls at a disadvantage. When different negativities are included in these difficulties, girls in rural areas face a disadvantage. The family order of disadvantaged girls and the educational or economic level of their parents (Abdullahi et al., 2013) also condemn them to some disadvantages and difficulties. In other words, the girls included in this study were from low-income families. According to Óskarsdóttir et al. (2016), they sometimes lack families and shelter. It should be noted that disadvantaged girls in the countryside, whose lives are almost on the edge, are also exposed to abuse (Óskarsdóttir et al., 2016). These issues render girls vulnerable and disadvantaged. The negative experiences can also manifest in an educational environment.

It should be noted that disadvantaged girls in rural areas are among the most vulnerable groups, especially in terms of access to education despite progress in primary education (Thamminaina et al., 2020; UNESCO, 2015). As stated above, female students in rural areas; These are children who have limited access to education, are forced into child marriage, have poor families, have low levels of parental education, are responsible for taking care of their siblings, and face early school leaving. Often their basic education is interrupted. In this respect, it is appropriate to say that disadvantaged girls in rural areas cannot complete primary or secondary schools (Abdullahi et al., 2013). In other words, girls in rural areas receive less than four years of education on average (Jain et al., 2017). However, according to Hallfors et al., (2011), disadvantaged girls must stay at school and show continuity. Staying in school motivates girls to work harder and brightens their future lives. Teachers have also played an important role in this regard. In particular, the presence and support of female teachers are effective in improving girls' school attendance (Chakaninka et al., 2012). However, failure may occur when these conditions are unfavorable.

2. Early school leaving and its results on disadvantaged girls

Early school leaving is inevitable when failure and other barriers are effective for disadvantaged girl students who continue their education in rural schools ((Huang et al., 2021). However, not all of these factors are limited. According to Davis (1989), increasing education costs may also

* Corresponding author.

E-mail addresses: yusufkiziltas@yyu.edu.tr (Y. Kızıltas), hamza84.hg@gmail.com (H. Gönülal).

<https://doi.org/10.1016/j.childyouth.2024.107801>

Received 19 March 2024; Received in revised form 5 July 2024; Accepted 9 July 2024

Available online 14 July 2024

0190-7409/© 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

cause children to withdraw from school. In these cases, girls are forced to leave school early. Thus, the girls are forced to stay at home. Ultimately, the process of becoming a housewife begins for disadvantaged girls. The process of being a housegirl also turns into housework between the ages of 5–14 (Pocock, 2019). According to Allais (2009), girls spend more than 28 h a week on housework, which is more important than education (Thamminaina et al., 2020). Rural girls also experience different jobs, in addition to domestic and housework (Aslan, 2021; ERG, 2014). Girls also function as seasonal agricultural workers. In some cases, workmanship can manifest in different areas (Madra, 2022). The process that turns into child labor then evolves into a different situation. At the end of the process, disadvantaged girls get married at an early age and turn into child brides (Jain et al., 2017). In other words, school-aged girls become miniature housewives.

3. Disadvantaged girl students in Türkiye

In rural areas of Türkiye, especially in the Northeast, Central, and Southeast Anatolian regions (Madra, 2022; STGM, 2020; UNICEF, 2015), many disadvantaged girls are in the age of compulsory education but cannot attend school (Adıgüzel, 2013). According to the UNICEF (2015) report, approximately one out of every three children in Türkiye is out of school. It was stated that among these, girls were concentrated. Similarly, many girls of compulsory school age must leave or are forced to early school leave (Aslan, 2021; Batrya, 2017). For many years, policies have been developed in Türkiye to ensure the attendance of girls in schools. Campaigns such as ‘Daddy, Send Me to School (Baba Beni Okula Gönder)’, ‘Come on Girls to School (Haydi Kızlar Okula)’, ‘Snowdrops Project (Kardelenler Projesi)’, and ‘Every Girl is a Star (Her Kızımız Bi Yıldız)’ are projects especially aimed at ensuring the continuity and schooling of girls (STGM, 2020). However, it is difficult to conclude that this goal has been achieved. In this perspective, schooling (ERG, 2014) is below the desired level because many disadvantaged girls cannot attend school. Many disadvantaged girls of school age are exposed to abuse (STGM, 2020); they are married at a young age (Aslan, 2021), and they have too early school leave, news in the press, and research. According to a report written by Madra (2022), the issues of child, early, and forced marriage among girls in rural areas are emphasized. There are several reasons for these negative results. It is possible to divide these schools into in-school and out-of-school. Low family income (ERG, 2014), distance from schools (Wei & Yang 2015), transportation education that it produces, low number of classrooms, excessive teacher mobility in rural schools (Kızıltaş, 2021), tribal and ultra-conservative family structures, girls’ othering, and distrust of girls (Aslan, 2021) are some of the reasons outside of school (Dilli, 2006; Tunç, 2009). However, there are internal reasons why disadvantaged girls fall behind and remain away from school. Teacher neglect, low teacher quality, negative teacher attitudes, missing lessons, teacher mobility (Huang et al., 2021; Kızıltaş, 2021; Rana & Das, 2004) and marginalization of girls are also effective (Beckford & Lekule, 2021).

Considering the literature, the number of studies investigating disadvantaged girl students in rural areas of Türkiye is insufficient to meet these expectations. Most studies have focused on levels other than basic education and have not sufficiently examined disadvantaged girls (Adıgüzel, 2013; Aslan, 2021; Çolak, 2020; Tunç, 2009). It should be noted that studies conducted in this perspective are in the international literature (Achilike & Achilike, 2016; Yassin, 2020). Studies examining disadvantaged girls, especially in the context of early school leaving, are limited (Huang et al., 2021; Osman Mohamed, 2016; Thamminaina et al., 2016). These limitations make this study meaningful. The fact that the number of girls attending secondary education is higher in rural areas of Türkiye (Aslan, 2021) makes it necessary to question the early school leaving risk of early school for basic education (primary school, secondary school). This study is interesting and important in this regard. In Türkiye, almost no studies have examined early school leaving among girls attending basic education, especially among disadvantaged girls.

This deficiency highlights the importance of the present study. Especially in the eastern regions of Türkiye, although girls continue their basic education, they may early school leave when they reach secondary education (ERG, 2014). One of the aims of this research is to reveal reasons for the risk of early school leaving among disadvantaged girls in rural areas who continue their basic education. In line with these purposes, answers to the following questions were sought.

- What variables best explain the early school leaving risk scores of disadvantaged girl students?
- Which variables explain the early school leaving risk scores of the disadvantaged girls?

4. Method

4.1. Study design

In this study, quantitative research methods were preferred. This study had a relational (correlational) nature. Relational studies can be defined as those in which the relationship between two or more variables is examined without intervening with these variables (Fraenkel et al., 2012). In this study, the relational research model was preferred because it aimed to determine the relationships between the variables that predict early school leaving.

4.2. Sample

The sample of the research was four primary schools 4th, and secondary schools 5th, 6th, 7th, and 8th-grade girls ($n = 515$). The study sample was determined using the criterion sampling method, which is a purposive sampling method. Therefore, various criteria were used to identify the samples. Girls attending certain grade levels, schools located in rural areas in the city center, and districts were the criteria considered. These regions have also experienced intense immigration. It is also possible that there are regions in which population mobility is intense. These points clearly demonstrate the characteristics of the study sample.

Primary education in Türkiye is four years. After four years, the students are transferred to secondary school. The secondary education period is four years. After secondary school, the high school education process continues for four years. However, there is a risk of early school departure during transitions between levels. A student who finishes the fourth grade of primary school may run early school leaving before entering secondary school (5th grade). Students starting in the fifth grade may also face this risk as they start a new level. Likewise, some students (8th grade) who have completed secondary school generally do not continue secondary education. Compulsory formal education in Türkiye has a system of $4 + 4 + 4$. So first there is a 4-year primary school education, then a 4-year secondary school education and finally a 4-year secondary education. Although education is compulsory, early school leaving is high, especially for girls. This risk starts at the end of primary school and continues until the end of secondary school. In high school education, this risk often results in early school leaving. Therefore, the rate of early school leaving in Türkiye is higher in high school education (Küçükarslan, 2019). However, there is a risk of leaving school early in primary and secondary schools. Therefore, disadvantaged rural girls attending primary education may be at risk.

4.3. Data collection tool and data collection process

Data on the risk of early school leaving of disadvantaged girl students were collected using the Early School Dropout Risk Scale developed by Sütçü (2015). Permission to apply the scale was obtained from the researchers. The Early School Dropout Risk Scale consists of 33 items and three sub-dimensions. The scale was prepared as a five-point Likert scale. The internal consistency coefficient of the ‘perception of failure’ dimension is 0.81. Cronbach’s alpha coefficient is 0.84, and the

construct reliability coefficient is 0.86. The other sub-dimension of the scale is 'silent behavior'. The internal consistency coefficient of the silent behavior dimension is 0.82. Cronbach's alpha coefficient is 0.85, and the construct reliability coefficient is 0.86. The internal consistency coefficient of the 'antisocial behavior' subscale is 0.87. Cronbach's alpha coefficient is 0.92, and the construct reliability coefficient is 0.91. A time of 10–15 min was given to complete the scale. The items were scored between 1 (strongly disagree) and 5 (strongly agree). The scale was applied by the researcher, and data were collected from the students on the same day. The negative statements were scored using reverse coding. To collect personal information from girl students, answers to several questions were sought. Class, income, parental education level, and number of siblings in the family were also included in the personal information form.

4.4. Data analysis

The data were analyzed using SPSS 25.0. Descriptive statistics and 'two-step clustering analysis,' which divided the heterogeneous sample into homogeneous subsets, were used in the data analysis. According to Kayri (2007), transforming heterogeneous data into homogeneous data allows analysis results to be healthier. In two-step clustering analysis, the silhouette coefficient was considered when determining the number of clusters. The silhouette coefficient has values between -1 and $+1$. As this coefficient approaches the value of $+1$, it means that the difference between clusters is minimal, and the difference between clusters is maximal (Rousseeuw, 1987). As a result of the two-step clustering analysis applied to the research data, the obtained silhouette coefficient was 0.70. This value indicates a good level of differentiation between and similarities within clusters. However, when the number of clusters was determined by the researcher (for example, when the number of clusters was chosen as two), the silhouette coefficient was found to be very low. In this respect, it is meaningful that there are three clusters and the silhouette coefficient is good.

After two-step clustering analysis, CHAID, a data-mining method, was applied to the data. In CHAID analysis, with a powerful iteration algorithm, the entire universe can be divided into stable subnodes. Doğan and Özdamar (2003) stated that it is a technique that repeatedly divides a population into different subgroups or segments such that the variation in the dependent variable is minimum within the sections and maximum between the sections. This process ensured normality and homogeneity of the data distribution. In addition, CHAID analysis allows continuous and categorical data to be modeled simultaneously. Similarly, it uses chi-square (χ^2) statistics, especially since it deals with the relationships and interactions of independent variables. In the CHAID analysis, the results of stepwise regression analysis performed in the first step were examined. As a result of this analysis, the variable with the highest F value among the independent variables, whose effect on the dependent variable is found to be statistically significant, takes the first place in the CHAID diagram (Kayri & Boysan, 2007). In this study, the risk of early school leaving before CHAID analysis was accepted as a dependent variable. The most influential variable on the risk of early school leaving was revealed. According to Yavuz (2020), the reasons for choosing CHAID analysis are its ability to benefit from large samples, which can be explained by providing reliable estimates, estimating lost values in independent variables, working with all kinds of variables, and being easy to interpret due to its visual appearance. Similarly, in CHAID analysis, not only the relationship between the dependent variable and the independent variable, but also the relationships of the independent variables with each other can be analyzed, and all types of variables can be included in the analysis at the same time as the reasons why CHAID analysis is preferred in this study.

The Cronbach's alpha coefficient of the whole early school leaving risk scale was 0.90. Considering the values on which Guilford is based on the interpretation of the scale's reliability (Putri et al., 2019), the scale is considered reliable. If the reliability coefficient is in the range of $0.80 \leq$

$\alpha < 1.00$, the scale is highly reliable.

4.5. Findings

Regarding the research data, frequency, percentage distributions, applied CHAID analysis, and descriptive statistics regarding some findings obtained from the scale are given below.

Table 1 presents the structure of the sample's variables. Notably, the total scores on the scale were categorical. This process was applied considering the possibility of heterogeneity among the girl students in the sample. Table 2 presents the results of the two-stage clustering analysis.

In the first cluster, there were 174 girl students (33.8 %) leaving high early school, with a total mean score of 151.26 ± 2.27 . In the second cluster, 170 girl students (33.0 %) were early school leavers and middle-level students, with an item total score of 127.06 ± 7.79 . In the third cluster, 171 girl students (33.2 %) had a low level of early school leaving, with an item total score of 95.78 ± 12.6 .

As a result of the CHAID analysis, this study aimed to reveal the relationship between the answers given by the girl students to the early school leaving risk scale and the students' grade levels, the number of siblings in the family, the education level of the mother, the education level of the father, and the income level. It is also important to reveal the number of students at risk and the dominant variables in these risks. The results of the CHAID analysis applied to all the early school leaving risk scale scores are shown in Fig. 1. The scores obtained from the early school leaving risk scale were used as dependent variables. The algorithm F test was used to determine the variables that affect the dependent variable and at which level the effect is higher.

According to the results of the analysis, the most dominant variable on the early school leaving risk of girl students is the number of siblings in the family ($\chi^2 = 167.295$, $p = 0.000$, $df = 4$). This variable divided the dependent variable into three groups and significant differences were observed between the groups. Of the students, 30.1 % had 4–6 siblings; and 15.3 % had students 0–3 siblings. Among the students, 54.6 % had seven or more siblings. 86.1 % of girl students with 0–3 siblings, that is, the vast majority, were not at risk of early school leaving. The proportion of students at high risk in this group was very low at only 5.1 %. In

Table 1
Structure of Variables Used in Chaid Analysis.

Variable Name	Value (modalities)	f	%	MS	IV/ DV
Grade Level	4th grade	80	15.5	NV	IV
	5th grade	100	19.4		
	6th grade	102	19.8		
	7th grade	101	19.6		
	8th grade	132	25.6		
Number of Siblings	0–3 sibling	79	15.3	NV	IV
	4–6 sibling	155	30.1		
	7, and above sibling	281	54.6		
Mother Education Level	Illiteracy	136	26.4	NV	IV
	Primary school	89	17.3		
	Secondary school	182	35.3		
	High school, and above	108	21.0		
Income Level	Low	298	57.9	NV	IV
	Moderate	191	37.1		
	High	26	5.0		
	Illiteracy	70	26.4		
Father Education Level	Primary school	136	17.3	NV	IV
	Secondary school	238	35.3		
	High school, and above	71	21.0		
School Early School Leaving Risk	High risk	174	33.8	NV	DV
	Moderate risk	170	33.0		
	No risk	171	33.2		

MS is Measurement Scales; NV is Nominal Variable; f is Frequency; % is Percentage; IV is Independent Variable; DV is Dependent Variable.

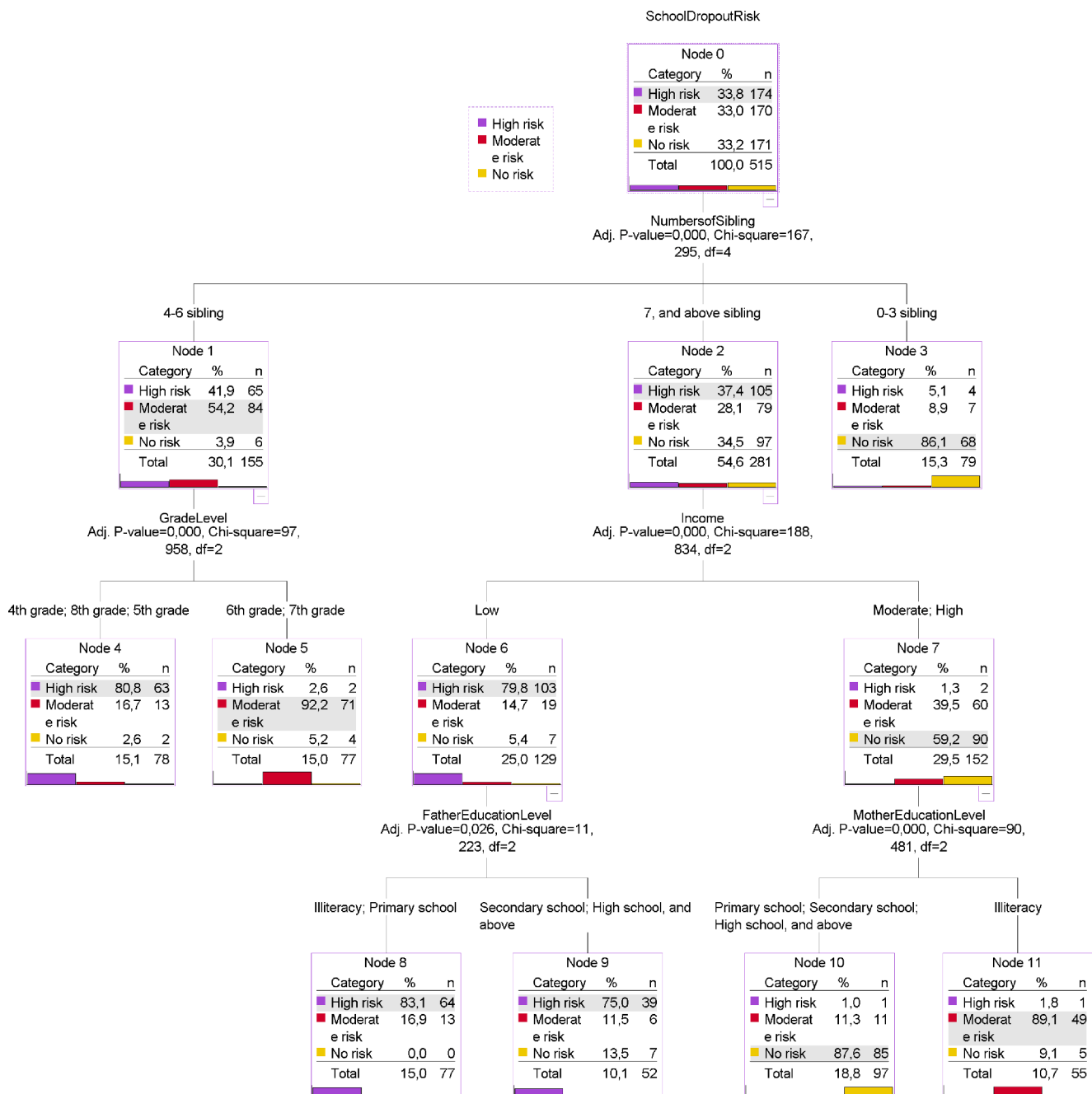
Table 2
Two-Step Clustering Analysis Results.

Cluster Number	Cluster Name	n	$\bar{X}$	%	ss
1	High risk	174	151.26	33.8	2.27
2	Moderate risk	170	127.06	33.0	7.79
3	No risk	171	95.78	33.2	12.6

other words, early school leaving was lower in girls with 0–3 siblings. In addition, 54.2 of the girls who had 0–4 siblings were at moderate risk of early school leaving. In addition, 41.9 % of girl students with 0–4 siblings were at high risk, which is very important. 37.4 % of girl students with seven or more siblings were at risk of early school leaving. In addition, the proportion of girls with seven or more siblings at moderate risk was 28.1 %. Among the girls who had 0–7 siblings, the proportion of

students who did not face early school leave was 34.5 %, which is very important.

The variable that best explains the cluster formed by girl students with to 4–6 siblings is the grade level variable ($\chi^2 = 97.958$; $p = 0.000$, $df = 2$). This variable divided the dependent variable into two groups and significant differences emerged between the groups. 50.32 % of female students with 4–6 siblings are in 4th grade, 5th grade and 8th grade; 49.68 % of them are 6th and 7th grade female students. Grade levels 4,5,8. In a large proportion of girl students, 80.8 % had a high risk of early school leaving; 16.7 % had a moderate risk of early school leaving. 4th,5th,8th. The percentage of those who did not have the risk of leaving school among those at the grade level was 2.6 %, which is quite low. In addition, grade levels of 6., 7. While a large proportion of girl students (92.2 %) had a high risk of early school leaving, 5.2 % did not have an early school leaving risk. 6.,7. The percentage of those with



a medium level of early school leaving risk among those at grade level was 2.6 %.

The variable that best explained the cluster formed by girl students with seven or more siblings was the income level ($\chi^2 = 188.834$; $p = 0.000$, $df = 2$). This variable divided the dependent variable into two groups and significant differences emerged between the groups. Of the girls, 45.90 % with seven or more siblings had low incomes, and 54.10 % had medium and high incomes. A large proportion of 'low' girl students, ie 79.8 %, are at high risk of early school leaving; 14.7 % have a moderate risk of early school leaving. The percentage of students who did not leave early school was low (5.4 %). On the other hand, a large proportion of middle- and high-income girl students (59.2 %) did not have a risk of early school leaving. However, 39.5 % of girls had a medium level of early school leaving risk. The rate of high early school leave among girl students at the middle- and high-income levels was 1.3 %.

The variable that best explained the cluster formed by low-income girls was the father's education level ($\chi^2 = 11.223$; $p = 0.026$, $df = 2$). This variable divided the dependent variable into two groups, and significant differences emerged between the groups. Of the fathers of low-income girls, 59.7 % were illiterate and 41.3 % were students with secondary school, high school, and higher education levels. There is a high risk of early school leaving in 83.1 % of girls whose fathers are at the primary school education level, illiterate, and 16.9 % at a medium level. There were no illiterate students whose fathers' education level was primary school and who were not at risk of early school leaving. Likewise, 75 % of girl students with a father's education level of secondary school, high school, or higher had a high risk of early school leaving, and 11.5 % had a moderate risk of early school leaving. On the other hand, 13.5 % of the girls whose fathers' education level was secondary school, high school, or higher did not have an early school leaving risk.

The variable that best explained the cluster formed by girl students with medium and high incomes was the mother's education level variable ($\chi^2 = 90.481$; $p = 0.000$, $df = 2$). This variable divided the dependent variable into two groups, and significant differences emerged between the groups. Of the mothers of middle- and high-income girls, 63.8 % were in primary school, secondary school, high school, and above; 36.2 % of them consisted of students with an education level of 'not being literate.' There was no risk of early school leaving in 87.6 % of girl students whose mothers' education level was primary, secondary, high school, or higher. By contrast, 11.3 % of the students had a medium level of early school leaving risk. Of the students whose mothers were illiterate, 89.1 % had a medium level of early school leaving. On the other hand, 9.1 % of students whose mothers were illiterate did not leave early school.

The classification process was not complete until all the performances were evaluated. Tables 3 and 4 provide the basic information on the performance of the CHAID model in terms of its accuracy and predictive potential.

Table 2 presents the estimation risk as a percentage of misclassified observations. Our findings show a 14.8 % risk of girl student misclassification in terms of the early school leaving risk scale (based on the whole sample) compared with 19.6 % when a test sample was used in model cross-validation. Table 3 presents the classification matrix containing the empirical and modeled values, namely the actual (observed) and predicted categories, according to the dependent variable categories. Thus, it can be concluded that the overall accuracy of the model is 85.2 %. A high value of is important. In other words, the model

Table 3
Risk.

Estimate	Cross-validation
Re-substitution	
0.148	0.196

Source: Authors' calculations.

Table 4
Classification Matrix.

Failure Perception		Predicted			% of correctly classified
		1.00	2.00	3.00	
Observed	1.00	166	3	5	95.4 %
	2.00	32	120	18	70.6 %
	3.00	9	9	153	89.5 %
Overall %		40.2 %	25.2 %	34.2 %	85.2 %

correctly classified 174 of 515 girls in the observed sample (the main diagonal of the matrix). Significant differences in classification accuracy were observed when the dependent variable categories were observed. However, the percentage structure of the modeled (predicted) values according to the categories of the dependent variable (40.2 %, 25.2 %, and 34.2 %) was not significantly different from that of the original data (33.8 %, 33.0 %, and 33.2 %) and faithfully represented the basic relationships between them.

Table 5 shows that the 8th node is the best terminal node that distinguishes girls who are at risk of early school leaving from those who are not at risk of early school leaving ($n = 64$, 36.7 %). This node is a cluster that includes 77 girls whose fathers are illiterate and are in primary school; 83.1 % of them are correctly classified. In this study, the gain values were examined to determine the second-best node, and the 4th node was found to be highly effective in identifying high-risk students in terms of early school leaving risk ($n = 63$, 36.2 %). This node is the cluster that is effective for the risk of early school leaving, in which 78 people had 0–3 siblings in the number of siblings variable, and 80.8 % of them were correctly classified. In addition, it was determined that the 10th node ($n = 1$, 0.57 %) was the least informative in distinguishing students according to their high risk of early school leaving. This node is the cluster in which 97 students whose mother's education level is primary school, secondary school, high school, and above are included, and they are classified correctly at the rate of 1.00 %.

5. Conclusions, and discussion

In Türkiye, early school leaving is more common in secondary education (Küçükarslan, 2019; MEB-UNICEF, 2013). According to the Eurostat (2017) report, Türkiye has the highest rate of leaving early school in Europe. It should be noted that early school leaving is higher for girls, especially in rural areas and the south-eastern and eastern regions of Türkiye. Sixty% of girls between the ages of 11 and 14 years were not enrolled in school (UNICEF, 2011). According to a MEB (2013) report, early school leaving is higher among girls, especially in secondary education. These realities make it necessary to examine the risk of early school leaving at primary and secondary school levels. The risk of early school may leave may start from primary and secondary school levels and result in early school leaving in secondary education institutions. According to the research results, although early school leaving is higher in secondary education institutions, the risks may start at the end of primary and secondary schools. Warning bells about early school leaving start to ring in primary school and especially in secondary

Table 5
Chaid Analysis Nodes Gain Values Table.

Node	Node		Earning		Answer
	n	%	n	%	
8	77	15.0	64	36.7	83.1
4	78	15.1	63	36.2	80.8
9	52	10.1	75	43.1	75.0
3	79	15.3	4	2.29	5.1
11	55	10.7	1	0.57	1.8
10	97	18.8	1	0.57	1.0

Dependent variable: Risk of early school leaving.

Target category: High risk.

school, and it can turn into direct action in secondary education, namely early school leaving. In addition, the results revealed that most students were in the high- and medium-level early school leaving risk groups.

According to the results of this study, a high number of siblings in the family (seven or more siblings) is the most important variable influencing the risk of early school leaving disadvantaged girls in primary and secondary schools. On the other hand, having 0–3 siblings is not a risk factor for early school leaving. In other words, while a decrease in the number of siblings in the family reduces the risk of early school leaving, an increase in the number of siblings increases the risk of early school leaving. In this sense, the number of siblings is an important predictive variable for early school leaving (Yorgun, 2014). It is worth discussing how a high number of siblings plays a role in the risk of early school leaving. The fact that the number of siblings in the family is high creates a traditional expectation for girls to take responsibility at home and care for siblings (Hekela, 2014). In other words, girls look after their younger siblings instead of being sent to school (Yavuz et al., 2016). Girls become almost defenseless and powerless in the face of these obligations. According to Canagarajah and Coulombe (1997), the number of younger siblings at the age of six may cause girls to have early school leaving risks. Finally, taking care of siblings and other household chores tires girls and causes absenteeism at school (Velasco, 2001). The age of siblings is as important as the number of siblings. As the age of siblings increases, the risk of early school leaving decreases, whereas the risk of early school leaving occurs as the age range of siblings decreases (Glick & Sahn, 2000). In other words, girls come to the aid of siblings, who lack self-care skills. Thus, in a sense, the role of miniature adults is assigned to disadvantaged girls who already have low socioeconomic and parental education levels. Tunç (2009) makes the analogy of 'second motherhood' for girls in Türkiye. In addition, making long-term investments in the family entirely for sons may push daughters into the background. In other words, boys, not girls, are the priority in the family's plans. It is preferred to send boys to school instead of girls (Achilike & Achilike, 2016). This may also be effective in the preference of girls for the role of sibling caretaker. Family resources were not evenly distributed among the children. In any case, the following are true: The large number of siblings in the family structure (Çolak, 2020) and polygamy in the family (Osman Mohamed 2006), the status of the girl as the elder sister in the family, and the responsibilities brought by this can cause early school leaving in the long run (Güngör, 2019). The mission imposed on girl children in domestic and family chores also negatively affects school attendance and ultimately paves the way for early school leaving risks (Hekela, 2014). According to Şimşek (2011), early school leaving risk increases as the number of siblings in the family increases in rural areas. In fact, according to Subrahmanyam (2016), one out of ten girls in rural areas faces early school leaving risk because they have to look after their siblings. In addition, girls who have to take care of their parents or siblings for various reasons and even lose one of their parents are vulnerable to exclusion from education (Hunt, 2008). In other words, not only the number of siblings, but also being responsible for taking care of siblings (due to illness) triggers early school leaving vulnerable girls.

Similarly, according to the results of this study, the high-grade level of disadvantaged girl students in rural areas is an important variable in the risk of early school leaving. As the grade level increases, the risk of early school leaving also increases (Admassie, 2003). While this risk was very high in the 4th, 5th, and 8th grades, it was moderate in the 6th and 7th grades. In other words, when starting secondary school and reaching the last year of secondary school, the risk of early school leaving reaches a high level. Gültekin and Anagün (2006) draw attention to this in their research. In contrast to the results of this research, 1–7. He states that early school leaving is relatively low among grades. However, he also draws attention to the fact that the probability of leaving school suddenly reaches 20 % in the last year of secondary school (8th grade). In a sense, risk gradually increases from the end of primary school to the end of secondary school. On the other hand, at the beginning of secondary

education, that is, in the 9th grade, the risk of early school leaving causes early school leaving. Early school leaving reaches its highest level (Küçükarslan, 2019; MEB, 2013). In other words, early school leaving and its risk at the end of secondary school continue, with early school leaving completely at the beginning of secondary education. Prakash et al. (2016) findings are also consistent with this situation. The risk is higher in transitions between the levels. 9th grade is a level where there is a transition between levels. This was the first grade of transition to secondary education. Similarly, primary school 4th grade is the last grade of primary school in Türkiye. Therefore, the 4th grade, which is the last class in primary school, or the 5th grade, which is the beginning of secondary school, also represents a transition between levels. In other words, according to the current research results, the risk of early school leaving is very high at these grade levels. Kosa and KinKesa (2015) pointed out that the risk of early school leaving is higher for girls at the end of secondary school. He states that an early school leaving risk after secondary school is also common. In other words, early school leaving is common when secondary education is initiated. He states that the most important reason for this situation is the family's socioeconomic status. Of course, the reasons are not limited to this. Early marriage also triggers this situation (Ahmadi, 2021; Duale et al., 2023). Undoubtedly, the onset of the risk of early school leaving at the end of primary school or the beginning of secondary school is an important problem. Şimşek (2011) also reveals the fact that the risk of early school leaving is high at the end of primary school in rural areas. On the other hand, Gökşen et al. (2005) emphasize that there is a more intense risk of early school leaving in middle school grades for girls. According to the results of this research, the risk of early school leaving is moderate in intermediate classes. According to Çalişıcı et al. (2023), some statements in research are also important. It has been stated that early school leaving risk or school dropout may occur in intermediate classes when academic failure (illiteracy) accompanies an increase in grade level at the secondary school level, especially in disadvantaged students. In other words, if girl students have had academic failure before, this problem may cause the risk of early school leaving after a certain period in the secondary school process.

According to another study, a low level of family income is more determinant in terms of the risk of early school leaving in basic education. Cardoso and Verner (2007) stated that a low-income level is the most effective factor in early school leaving. In addition, it has been revealed as a result of the present research that the middle and high levels of income in the family are not very effective in terms of the early school leaving risk of girl students. In other words, if income rises, the risk of early schooling is low at the basic education level, while low family income increases the risk of early school leaving (Peraita & Pastor, 2000). Income level determines the welfare and future of girls rather than that of boys (Foster, 1995). As such, a low income may play a greater role in girls' risk of early school leaving. This is also consistent with various research results that income is a determinant of girls' risk of early school leaving and whether they attend school (Dilli, 2006; Muia, 2015; Tunç, 2009). This is because girls from low-income families experience problems accessing educational materials. Parents who have difficulty meeting the food expenses of their daughters are reluctant to send their children to school. According to Rumberger and Lim (2008), the higher the class level, the higher the education costs. Thus, pressure on the child arises that the child should earn money. According to Kaya (2022), since the high number of children in the family also affects the income distribution and level of the family, these problems create an obstacle for girls not to go to school. Undoubtedly, these problems are also determinative of the risk of early school leaving (Yassin, 2020). Similarly, Gubert and Robilliard (2006) state that girls from low-income and income-shock families are vulnerable and face the risk of early school leaving. What is meant by Income shocks refer to the sudden unemployment of one or both parents and a sudden decrease in family income sources. On the other hand, as a result of the research, a high level of income also minimized the risk of early school leaving.

However, contrary to the results of this research, according to Çolak (2020), a high level of income may cause early school leaving. In addition, he states that even if families have a good income, they do not give importance to girls' education for various reasons, including traditional perspectives (Kaya, 2022), and security concerns. However, according to White and Kelly (2010), the risk of early school leaving is 10 times higher in children from low-income families than in those from high-income families.

According to another study, a low mother education level is a long-term determinant of the risk of early school leaving (Şimşek, 2011; Yorgun, 2014), although not directly. In this context, Gökşen et al. (2005) stated that 60 % of girls attend school when their mother's education level is good. When the mother's education level is primary school, the probability of leaving school is 85 %. The higher the mother's education level, the higher the probability of girls attending school (Çolak, 2020). In addition, Kumar et al. (2023) point out that mother's education increases the interaction between children and is decisive in being a role model, and states that this reduces the risk of early school leaving. The educated mother was conscious of this. The lack of a conscious mother may pose a risk of early school leaving (Tunç, 2009). As Yasin and Aslam (2018) stated, mothers with a certain level of education can be an important force against fathers in helping girls continue and supporting their education. Thus, when mothers are strong in their patriarchal structures when compared with fathers, they can prevent early school leaving risk girls (Shahidul, 2012). On the other hand, sometimes the mother's job can cause problems for the girls. The fact that the mother works in a job may cause the daughter to take responsibility for the family (Fuller & Liang, 1999). This responsibility can also trigger early school leaving. However, according to this theory, the fact that the mother is employed in the formal economy (an insured job) and the increase in the level of education may also be determinative in the schooling of girls (Hoşgör & Smits, 2006). The mother is an important variable because sometimes even having a say in the family can have an impact on the fate of girls. Having a say in the mother can create effective results in preventing early school leaving girls (Shahidul, 2002). On the other hand, it is necessary to look at how important the level of mother and mother's education is from other perspectives, which are different from the results of this study. Even when mothers' education and awareness levels are low, this deficiency may be effective in girls' risk of early school leaving. In such a case, according to families, daughters will marry and work for their husbands when the time comes. This is why girls are guests, and it does not make sense to invest in girls by sending them to school. It is possible to see this, especially in parents with low educational levels. This assumption between mothers and fathers imposes a cycle of discrimination on girls' schooling. However, when one parent goes to school or the mother is literate, parents are more likely to want to send their daughters to school (Khan et al., 2011).

According to another result of the research, the low educational level of the father, that is, the father's illiteracy or being a primary school graduate, is a long-term determinant of the risk of early school leaving of disadvantaged girls students. This finding is consistent with the results of various studies (Sajjad et al., 2012; Şimşek, 2011). According to Shahidul (2012), the father is an important figure, particularly in male-dominated societies. Decisions regarding the future of girls and their school attendance are also important. The educational investments of fathers in rural areas are generally directed towards their sons (Tunç, 2009). Even fathers do not want girls of a certain age to attend school. Fathers prefer to marry their daughters at a young age (Çolak, 2020). Therefore, this situation reveals the importance and need for fathers to be conscious and have a high level of education. No et al. (2012) found that the education level of the father, rather than that of the mother, is more effective in the attendance of girls at school. In other words, the higher the education level of the father, the higher the child's chance of staying at school and the lower the risk of early school leaving. According to Holmes (2003), the education level of the mother increases

the schooling of girls, and the education level of the father increases the length of stay in school. In other words, an increase in the education level of fathers reduces the risk of early school leaving (Gouda & Sekher, 2014). The low educational level of both mothers and fathers is effective in reducing the risk of early school leaving. According to Yorgun (2014), a low level of education may decrease interest in school and the value given to education within a family. Thus, follow-up of children at school can be neglected. Support for motivating children for education may also be lacking. Consequently, there may be a risk of early school leaving.

5.1. Recommendations of the study

Many projects have been conducted in Türkiye to ensure the attendance of girls in rural areas at school. These projects have names such as 'Snowdrops,' 'Come on Girls to School,' 'Daddy Send Me to School,' 'Mother-Daughter Are at School,' and 'Every Girl Is a Star.' The results of these projects should follow. The situation of girls enrolled in schools as a result of these projects should also be examined. Disadvantaged girls should be regularly provided stationery and financial support. It should be noted that this support was spent on the child by the family. Visits should be made to families with constantly absent students, especially under the leadership of female teachers. Researchers can investigate the problem of child brides in girls who early school leaving risk as a result of early school leaving.

6. Limitations of the study

The data of the study are limited to girl students who continue their education in all classes of primary, senior, and secondary schools.

7. Ethics document of the study

The ethical approval certificate for this study was issued on 07/08/2023. It was taken with the permission of the Social and Human Sciences Publication Ethics Committee of the University of Van Yuzuncu Yil University numbered 2023/21.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Abdullahi, I., Zainalaludin, Z., & Paim, L. B. (2013). Empowering rural girls through education: Way of reducing vulnerability and improving family economic wellbeing in rural Northern Nigeria. *IOSR Journal of Humanities and Social Science (IOSR-JHSS)*, 18(1), 57–61.
- Achilike, B. A., & Achilike, C. C. (2016). Causes of Female Students Dropout in Secondary schools in Ebonyi state south east Nigeria: Psychological implications. *Journal of Educational Policy and Entrepreneurial Research (JEPER)*, 3(4), 103–110.
- Adigüzel, A. (2013). Schooling of Girl Children Barriers and Suggestions for Solution a Case Study of Sanliurfa. *EKEV Academy Journal*, 17(56), 325–344. <https://www.acarindex.com/dosyalar/makale/acarindex-1423877811.pdf>.
- Admassie, A. (2003). Child labour and schooling in the context of a subsistence rural economy: Can they be compatible? *International Journal of Educational Development*, 23(2), 167–185. [https://doi.org/10.1016/S0738-0593\(02\)00012-3](https://doi.org/10.1016/S0738-0593(02)00012-3)
- Ahmadi, A. (2021). Reducing school dropout rates among Afghan Girls. Friedrich-Ebert-Stiftung: <https://afghanistan.fes.de/e/reducing-school-dropout-rates-among-afghan-girls>.
- Allais, F. B. (2009). *Assessing the gender gap: Evidence from SIMPOC surveys*. Geneva: International Labour Organization.
- Aslan, G. (2021). Determinants of access to education: Factors preventing girls from being sent to upper-secondary education in rural areas. *Education and Science*, 46, 169–201. <https://doi.org/10.15390/EB.2021.9045>.

- Bashir, A., Khan, N., & Vaida, N. (2014). During the education process; Early marriage, being busy with housework, supporting the care and education of their siblings, lack of interest in classes, having to work in income-generating activities, and physical difficulties are also the difficulties. *Journal of Scientific and Innovative Research*, 3(3), 295–298.
- Batrya, A. (2017). *Türkiye’de cinsiyete dayalı başarı farkı (Gender-based achievement gap in Turkey)*. İstanbul: Eğitim Reformu Girişimi, Aydın Doğan Vakfı. https://www.aydindoganvakfi.org.tr/media/byseyyfb/pisa_tr.pdf.
- Beckford, C., & Lekule, C. (2021). Vulnerability, marginality and barriers to girls’ education: A preliminary exploration of the case of “house girls” in Tanzania. *The International Journal of Social and Development Concerns (IJSDC)*, 14(4), 48–64.
- Canagarajah, S., & Coulombe, H. (1997). *Child labor and schooling in Ghana*. Washington DC: World Bank.
- Cardoso, A. R., & Verner, D. (2007). *School dropout and push-out factors in Brazil: The role of early parenthood, child labor and poverty*. Bon: Institute for study of labor (IZA).
- Chakaninka, W. W., Sichula, N., Sumbwa, P. I., & Nduna, M. (2012). The challenges of rural education in Africa. *South Africa Rural Educator*, 2, 6–12.
- Çalışıcı, M., Yüksel, D., & Yağır, E. (2023). Opinions of teachers working in schools in disadvantaged regions on existing problems. *International Journal of Social Sciences and Humanities Sciences Research (JSHSR)*, 10(92), 486–499. <http://doi.org/10.26450/jshsr.3490>.
- Çolak, İ. (2020). *The reasons of women’s school leaving and views on the education of girls*. Tokat: Tokat Gaziosmanpaşa University. Unpublished master thesis.
- Davis, D. (1989). Chinese social welfare: Policies and outcome. *The China Quarterly*, 119, 577–597. <https://doi.org/10.1017/S0305741000022943>
- Dilli, C. (2006). *Reasons why girls of compulsory education age do not go to school (Unpublished master thesis)*. Elazığ: Fırat University.
- Doğan, N., & Özdamar, K. (2003). CHAID analysis and an application related with family planning. *T Klin J Med Sci*, 23(5), 392–398.
- Duale, H. A., Aden, M. Y., Mohamed, J. O., & Ismail, A. A. (2023). Factors affecting female student dropout rates in Hargeisa, Somaliland high schools. *Research Square*, 1. <http://doi.org/10.21203/rs.3.rs-3143590/v1>.
- Erg. (2014). *Türkiye eğitim sisteminde eşitlik ve akademik başarı: Araştırma raporu ve analiz*. İstanbul: Açık Toplum Vakfı, Eğitim Reformu Girişimi. Retrieved from <http://kasauu.ankara.edu.tr/wp-content/uploads/sites/34/2013/02/ERGesitlikWEB.22.05.14.pdf>.
- Eurostat. (2017). *Key figures on Europe*. Luxembourg: Publications Office of the European Union. Retrieved from <https://ec.europa.eu/eurostat/documents/3217494/8309812/KS-EI-17-001-EN-N.pdf/b7df53f5-4faf-48a6-aca1-c650d40c9239>.
- Foster, A. (1995). Prices, credit constraints and child growth in low-income rural areas. *The Economic Journal*, 105(430), 551–570. <https://doi.org/10.2307/235017>
- Fraenkel, J., Wallen, N., & Hyun, H. H. (2012). *How to design and evaluate research in education*. Boston: McGraw Hill.
- Fuller, B., & Liang, X. (1999). *Which girls stay in school? The influence of family economy, social demands, and ethnicity in South Africa*. Washington, DC: National Academy Press.
- Glick, P., & Sahn, D. E. (2000). Schooling of girls and boys in a West African country: The effects of parental education, income, and household structure. *Economics of Education Review*, 19(1), 63–87.
- Gouda, S. M., & Sekher, T. V. (2014). Factors leading to school dropouts in India: An analysis of national family health survey-3 data. *IOSR Journal of Research & Method in Education (IOSR-JRME)*, 4(6), 75–83.
- Gökşen, F., Cemalçılar, Z., Gürlelel, C., & F., . (2005). *Research on policies aimed at school dropout and its monitoring and prevention in primary schools in Turkey (Türkiye’deki ilköğretim okullarında okulu terk ve izlenmesi ile önlenmesine yönelik politikalar araştırması)*. İstanbul: ERG, AÇEV.
- Gubert, F., & Robilliard, A. (2006). *Do household income shocks affect school attendance in rural areas?: A case study of Madagascar*. Development, Institutions & Analyses de Long terme: Working Paper. Paris.
- Gulma, A. M., & Lawal, F. G. (2010). Issues and challenges to women education in Nigeria. *Sardauna Journal of Multidisciplinary Studies*, 1(2), 172–178.
- Gültekin, M., & Anagün, Ş. S. (2006). The condition of Turkish education system from the aspect of European union’s area and indicators determining the quality in education. *Journal of Social Sciences*, 2, 146–170.
- Güngör, G. (2019). *The reasons and solution suggestions of school drop out in vocational and technical secondary education: A holistic program recommendation (Unpublished doctoral thesis)*. Kocaeli: Kocaeli University.
- Hallfors, D., Cho, H., Rusakaniko, S., Iritani, B., Mapfuno, J., & Helsen, C. (2011). Supporting adolescent orphan girls to stay in school as HIV risk prevention: Evidence from a randomized controlled trial in Zimbabwe. *American Journal of Public Health*, 101(6), 1082–1088. <https://doi.org/10.2105/AJPH.2010.300042>
- Hekela, L. P. (2014). *Exploring factors contributing to female students drop out rates in community secondary schools in Tanzania: A case of Mbinga district (Unpublished master thesis)*. Tanzania: University of Tanzania.
- Holmes, J. (2003). Measuring the determinants of school completion in Pakistan: Analysis of censoring and selection bias. *Economics of Education Review*, 22(3), 249–261. [https://doi.org/10.1016/S0272-7757\(02\)00024-9](https://doi.org/10.1016/S0272-7757(02)00024-9)
- Hoşgör, A. G., & Smits, J. (2006). *The status of rural women in Turkey: What is the role of regional differences*. <https://jeroensmits.info/assets/2007/01/Gunduz-Hosgor-SmitsRuralWomenTurkey.pdf>.
- Huang, J., Huang, Y., Lin, H., & Yang, Z. (2021). Chinese rural female students’ situation in accessing to education resources A literature review on Chinese urban-rural differences. *Advances in Social Science, Education and Humanities Research*, 586, 128–134.
- Hunt, F. (2008). ‘Dropping out from school: A cross-country review of literature’. *CREATE Pathways to Access No 16*. University of Sussex.
- Jain, P., Agarwal, R., Illaiy, R., & Devi, J. (2017). Women education in rural India. *International Journal of Social Sciences and Humanities*, 1(1), 21–26. <http://doi.org/10.29332/ijssh.v1n1.12>.
- Kaya, H. (2022). Reasons for girls not attending formal secondary education in rural areas, according to the opinions of parents and students (Kırsal kesimde kız çocuklarının veli ve öğrenci görüşlerine göre örgün ortaöğretime devam etmeme sebepleri). Retrieved from *International Social Sciences*, 8(104), 4011–4025 <https://sssjournal.kl.com/files/sssjournal/e26b7ca1-d463-4356-8bd4-86cdab7ad23f.pdf>.
- Kayri, M. (2007). Two-step cluster analysis and its application in research. *Eurasian Journal of Educational Research*, 28, 89–99.
- Kayri, M., & Boysan, M. (2007). Using Chaid analysis in researches and an application pertaining to coping strategies. *Ankara University, Journal of Faculty of Educational Sciences*, 40(2), 133–149.
- Khan, G. A., Azhar, M., & Shah, S. A. (2011). Causes of primary school drop out among rural girls in Pakistan. Retrieved from *Sustainable Development Policy Institute (SDPI)*, 114, 1–8 <https://sdpi.org/sdpiweb/publications/files/W-119.pdf>.
- Kızıltaş, Y. (2021). Orantısız öğretmen hareketliliği: İki dilli öğrencilerin ağırlıkta olduğu kırsal bölgelerdeki okullarda yaşanan bir sorunun anatomisi (Disproportionate teacher mobility: Anatomy of a problem in rural areas schools with a predominance of bilingual students). Kocaeli University. *Journal of Education*, 4(2), 440–475. <https://doi.org/10.33400/kuje.956378>
- Kosa, B., & KinKesa, K. (2015). *Determinants of student dropout in Cambodia’s primary and lower secondary schools: A survey of program interventions*. Kirkusreviews.com/book-reviews/tish-rabe/on-the-first-day-of-kindergarten/.
- Kumar, P., Patel, S. K., Debbarma, S., & Saggurti, N. (2023). Determinants of School dropouts among adolescents: Evidence from a longitudinal study in India. *PLoS ONE*, 18(3), 1–17. <https://doi.org/10.1371/journal.pone.0282468>
- Küçükarslan, E. (2019). *Executives, teachers and students’ views on the factors and prevention strategies that affect vocational high school students’ dropouts: Comparison with EU countries (Unpublished master thesis)*. Ankara: Ankara University.
- Madra, A. (2022). Girls’ education in Turkey: Obstacles, interventions and opportunities (Türkiye’de kız çocukların eğitimi: Engeller, müdahaleler ve olanaklar). Retrieved from <https://sunaninkizlari.org/Raporlar/turkiyede-kiz-cocuklarin-egitimi.pdf>: Eğitim Reformu Girişimi, Suna’nın Kızları.
- Meb. (2013). *Secondary Education Monitoring and Evaluation Report (Ortaöğretim İzleme ve değerlendirme raporu)*. MEB Ortaöğretim Genel Müdürlüğü: Ankara.
- Meb-unicef. (2013). *Policy recommendations report on grade repetition in secondary education, reasons for school dropout and children excluded from formal education (Ortaöğretimde sınıf tekrarı, okul terk sebepleri ve örgün eğitim dışında kalan çocuklar politika önerileri raporu)*. Ankara: Aydoğdu Ofset Matbaacılık.
- Muia, M. M. (2015). Home based factors contributing to drop out of girls in mixed day secondary schools in mito-andei division kibwezi sub-county makueni county (Unpublished master thesis). South Eastern Kenya: South Eastern Kenya University.
- No, F., Sam, C., & Hirakawa, Y. (2012). Revisiting primary school dropout in rural Cambodia. *Asia Pacific Education Review*, 13, 573–581. <https://doi.org/10.1007/s12564-012-9220-2>
- Óskarsdóttir, S. S., Baldursdóttir, S., & Einarisdóttir, J. (2016). ‘Vulnerable’ girls, education and non-governmental organisation sponsorship programmes in Kampala. *Childhood*, 23(3), 410–422. <https://doi.org/10.1177/0907568216631027>
- Osman Mohamed, M. M. (2016). *Factors influencing female students’ dropout rate in secondary schools in Mogadishu, Somalia (Unpublished master thesis)*. Nairobi: Nairobi of University.
- Peraita, C., & Pastor, M. (2000). The primary school dropout in Spain: The influence of family background and labor market conditions. *Education Economics*, 8(2), 157–168.
- Pocock, N. S. (2019). *Child domestic worker prevalence in Myanmar and Southeast Asia: Briefing note*. London, UK: London School of Hygiene and Tropical Medicine.
- Prakash, R., Bhattacharjee, P., Thalinja, R., & Isac, S. (2016). *Regional, social and gender disparities in secondary school attainment in Karnataka, South India*. Rajajinagar, Bengaluru: Karnataka Health Promotion Trust. <http://doi.org/10.13140/RG.2.2.32262.78402>.
- Radiwala, A., & Molwane, M. S. (2021). A study on the challenges faced by rural women in accessing education. *Journal of Scientific Research*, 65(4), 13–17. <http://doi.org/10.37398/JSR.2021.650402.13-67>.
- Rousseuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.
- Putri, H. E., Isrokatun, I., Majid, N. W., & Ridwan, T. (2019). Spatial sense instrument for prospective elementary school student. *Journal of Physics: Conference Series*, 1318(1), 1–7. <https://doi.org/10.1088/1742-6596/1318/1/012142>
- Rana, K., & Das, S. (2004). Primary education in Jharkhand. *Economic and Political Weekly*, 39(11), 1171–1178.
- Rumberger, R. W., & Lim, S. A. (2008). *Why students drop out of school: A review of 25 years of research*. Santa Barbara, CA: California Dropout Research Project.
- Sajjad, H., Iqbal, M., Siddiqui, M. A., & Siddiqui, L. (2012). Socio-economic determinants of primary school dropout: Evidence from South East Delhi. *India. European Journal of Social Sciences*, 30(3), 391–399.
- Shahidul, S. M. (2002). Marriage market and an effect on girls’ school dropout in Bangladesh. *Journal of Alternative Perspectives in the Social Sciences*, 4(2), 552–564.
- Shahidul, S. M. (2012). Household decision-making process: It’s effect on school dropout behavior for girls in the secondary school level in Bangladesh. *International Education Studies*, 6(1), 132–141. <https://doi.org/10.5539/ies.v6n1p132>
- Stgm. (2020). *Monitoring the access of children at risk to their right to education (Risk altındaki çocukların eğitim hakkına erişiminin izlenmesi)*. İzmir: Sürdürülebilir Kalkınma ve Girişimcilik Derneği. İzmir Çocuk Çalışmaları Ağı Retrieved from <https://www.stgm.org.tr/sites/default/files/2020-11/egitime-erisim-raporu.pdf>.

- Subrahmanyam, G. (2016). *Gender perspectives on causes and effects of school dropouts*. Stockholm. <https://cdn.sida.se/publications/files/sida62010en-gender-perspectives-on-causes-and-effects-of-school-dropouts.pdf>. Sida.
- Sütçü, Z. (2015). *Development of school dropout risk scale (Unpublished master thesis)*. Ankara: Gazi University.
- Şimşek, H. (2011). Dropout tendency among high school students and its reasons. *Journal of Educational Sciences Research*, 1(2), 27–47.
- Thamminaina, A., Kanungo, P., & Mohanty, S. (2020). Barriers, opportunities, and enablers to educate girls from Particularly Vulnerable Tribal Groups (PVTGs): A systematic review of literature. *Children and Youth Services Review*, 118, 1–10. <https://doi.org/10.1016/j.childyouth.2020.105350>
- Tunç, A.İ. (2009). Reasons why not go to school for girls Van sample. Retrieved from Yüzüncü Yıl University, *Journal of the Faculty of Education*, VI, 1, 237–269 <https://dergipark.org.tr/tr/download/article-file/146304>.
- Unesco. (2015). *Global monitoring report. Education for all 2000–2015: Achievements and challenges*. Paris: UNESCO Publishing.
- Unicef. (2011). *Haydi kızlar okula: Kız çocuklarının okullaşmasına destek kampanyası*. Ankara: Milli Eğitim Bakanlığı İlköğretim Genel Müdürlüğü. Retrieved from.
- Unicef. (2015). *Okul dışındaki çocuklar kişisel gelişim: Türkiye ülke raporu*. Unicef.org.
- Velasco, E. (2001). Why are girls not in school? Perceptions, realities and contradictions in changing Cambodia. <http://119.82.251.165:8080/xmlui/bitstream/handle/123456789/183/Why%20are%20Girls%20Not%20in%20School%20Perceptions,%20Realities%20and%20Contradictions%20in%20Changing%20Cambodia.pdf?sequence=1>.
- Wei, X., & Yang, Z. (2015). Distribution and supply of educational resources for Chinese residents: from macro to micro inequality. *Education academic monthly*, 01, 50-58. <http://doi.org/10.16477/j.cnki.issn1674-2311.2015.01.005>.
- White, S. W., & Kelly, F. D. (2010). The school counselor's role in school dropout prevention. *Journal of Counseling & Development*, 88(2), 227–235.
- Yasin, S. A., & Aslam, S. (2018). *School dropout of rural girls in Pakistan: Exploring the role of gender discrimination* (Cilt 12). Journal of Research and Reflections in Education: 1-10. Journal of Research and Reflections in Education.
- Yassin, A. (2020). Research title: Major causes of girls' school dropout from general primary schools and related challenges in South Wollo and Oromia zones in Amahara region, Ethiopia. *Journal of Education and Practice*, 11(10), 34.42. <https://doi.org/10.7176/JEP/11-10-03>
- Yavuz, F. (2020). *Investigation of the attitudes and knowledge of primary school teaching mathematics teacher candidates (Unpublishes master thesis)*. Karaman: Karamanoğlu Mehmetbey University.
- Yavuz, M., Özkara, T., & Yıldız, D. (2016). The factors that cause female students to quit their formal education. Retrieved from *Education and Society*, 5(14), 271–273 <https://dergipark.org.tr/en/download/article-file/367711>.
- Yorğun, A. (2014). *Investigation of school drop out among high school students (Unpublished doctoral thesis)*. Ankara: Hacettepe University.